Faculty of Engineering and Information Technology
School of Computer Science

43008: Reinforcement Learning

Spring 2026

ASSIGNMENT-2 SPECIFICATION

Due date	Friday 11:59pm, **16th October 2026**
Demonstrations	If required.
Marks	30% of the total marks for this subject
Submission	1. A report in PDF (10-pages max) (10%) 2. Google Colab/iPython notebooks (25%)
Submit to	Canvas assignment submission

Note: This assignment is an individual work.

Summary

Assignment 2 requires students to demonstrate a firm understanding of solving Reinforcement Learning problems using Temporal Difference learning, DQN, and Policy Search techniques.

This involves formulating the given scenarios as RL problem, implement algorithms such as Q-Learning, DQN and Policy Search techniques to solve them. This assignment encourages students to explore RL algorithm design, customisation, and performance evaluation.

Assignment Objectives

The purpose of this assignment is to demonstrate competence in the following skills.

1. **Demonstrate Proficiency in RL Algorithms:** Students are expected to showcase their competence in designing and implementing reinforcement learning algorithms like Q-learning, DQN, and policy search techniques.
2. **Application to Problems:** Students are required to apply these algorithms to solve practical problems, reflecting the translation of RL concepts into practical solutions.

Tasks:

Description:

1. Apply TD method (e.g. Q-Learning), Deep Q-Learning and Policy Gradient techniques to solve the appropriate environment/scenarios.
2. Select appropriate environments/scenarios from **List-1** and **List-2** to apply Deep Q-Learning and Policy Gradient techniques to solve them.
3. Use a Temporal Difference Method such as Q-Learning to solve the compulsory environment/scenario.
4. For this assignment, students will select two scenarios from the given **List 1** and **List 2**, one each from the lists and a compulsory scenario/environment (in total 3 scenarios).

Scenario/Environment Selection:

1. Students should choose **two** scenarios from the following options, one from each list:
    a. **List-1: Open AI Gym/Gymnasium Atari Environment (select one**):
        i. Beam Rider
        ii. Breakout
        iii. Seaquest
        iv. Space Invaders
    b. List-2: Open AI Gym/Gymnasium Classic Control (select one):
        i. CartPole
        ii. MountainCar
        iii. Acrobot
2. **Mandatory Scenario/Environment** (for TD Based Method): Pendulum

Reinforcement Learning Methods:

1. Students must apply DQN and a Policy Gradient method for both selected scenarios from List-1 and List-2. **[4 experiments]**
    a. Students are required to create their own Custom Feature Extractor (CNN/ANN) if using Stable Baselines Library.
    b. Student can also use custom implementation of DQN and Policy Gradient methods.
2. For the mandatory scenario, students can select any one Temporal Difference Method (e.g. Q-learning). **[1 experiment]**

In summary, experiments should encompass two implementations with DQN, two experiments with Policy Gradient method and one with Temporal Difference method (mandatory scenario).

Task Execution for Each Scenario:

For each of the two selected scenarios, follow these steps:

1. **Problem Formulation:**
    - Describe the chosen scenario in detail.
    - Specify the state space, action space, and reward structure according to the scenario's characteristics.

2. **Method Implementation:**
    - Depending on the chosen reinforcement learning method (Temporal Difference Methods, DQN, or Policy Search Methods), implement the selected method to train an agent for the scenario.
    - Implement the chosen method's algorithms, parameters, and any necessary modifications.

3. **Training and Simulation:**
    - Train the agent using the chosen method, allowing it to learn an optimal policy or strategy over multiple iterations or episodes.
    - The agent must be trained for an adequate number of episodes/iterations to demonstrate meaningful learning and stable performance. Training for only a small number of iterations (e.g., 10–20) is not sufficient and will not be accepted.
    - As a minimum, students must use **1,000 training episodes for Classic Control/Pendulum environments and 1,000,000 for Atari environments.** Students may use additional training where required. Training progress must be demonstrated using appropriate learning curves and performance metrics.
    - Simulate the scenario using the trained agent and record relevant data, such as performance metrics, convergence times, or any other observations.

4. **Performance Evaluation:**
    - Evaluate the performance of the trained agent for each scenario.
    - Measure relevant performance metrics specific to each reinforcement learning method and scenario, e.g., success rate, average reward, or any scenario-specific criteria.

5. **Simulation and Discussion:**
    - Simulate the environments using the best policy/strategy.
    - Compare the strengths and weaknesses of TD Methods, DQN, and Policy Search Methods within the context of the specific scenarios.
    - Discuss any insights gained from the tasks.

Students are encouraged to thoughtfully select scenarios and reinforcement learning methods that align with their interests and expertise. This assignment

provides an opportunity to apply reinforcement learning techniques to a diverse set of scenarios and gain practical experience in solving complex problems.

Report Structure:

The report should include the following sections (template provided on Canvas):

1. **Introduction:** Provide a concise overview of the report, explaining the scenarios, and the algorithms used for the experiments.

2. **Scenario and Algorithm:** For each scenario selected elaborate on the states, actions, rewards, and goal/aim. For each algorithm selected, elaborate on the specifics, use-case, rationale, and general information about the algorithm.

3. **Diagram/Pseudocode/Flowchart:** Provide a generic flowchart/diagram for each of the technique being used. Provide pseudocode(s) for each algorithm you're implementing. Provide network diagrams illustrating any neural network(s) you are using for the network-based methods if selected.

4. **Experimental results and discussion:**
    a. **Experimental settings:** Provide information about the specific parameters and settings used for the scenario and algorithm.
    b. **Experimental Results:**
        i. **Training Details:** Discuss the training specifics, plots, and observations.
        ii. **Performance Evaluation:** Measure relevant performance metrics specific to each reinforcement learning method and scenario, e.g., success rate, average reward, or any scenario-specific criteria and observations.
        iii. **Discussion:** Share your insights on the strengths and limitations of each algorithm, based on the results derived from the two chosen scenarios. Share picture and plots showcasing simulation and results of the algorithm's result.

5. **Conclusion:** Provide a short paragraph detailing your understanding on the experiments and results.

Deliverables:

1. **Project Report** (maximum of 10 pages).
2. **Code notebook(s)** [in ipynb/colab format]. The submitted notebook(s) should **display the output after the execution of each code cell**.
3. Use appropriate comments and variable naming convention to make the code more readable.
4. Two separate iPython/Colab notebooks can be used, one for each scenario.

Assessment Submission:

Submission of your assignment is in two parts. You must upload a zip file of the iPython/Colab notebooks and the Report separately on Canvas. This must be done by the Due Date. You may submit as many times as you like until the due date. The final submission you make is the one that will be marked. If you have not uploaded your submission by the Due Date, or it cannot be run/tested, then your assignment will receive a zero mark. Additionally, the result/accuracies achieved and shown in the iPython/Colab notebooks should match the report. Penalties apply if there are inconsistencies in the experimental results and the report.

NOTE 1: It is your responsibility to make sure you have thoroughly tested your program to make sure it is working correctly.

NOTE 2: Your final submission to Canvas is the one that is marked. It does not matter if earlier submissions were working; they will be ignored. Download your submission from Canvas and test it thoroughly in your assigned laboratory session.

Return of Assessed Assignment:

It is expected that marks will be made available 2 weeks after the submission via Canvas.

Queries:

If you have a problem such as illness which will affect your assignment submission contact the subject coordinator as soon as possible.

Dr. Nabin Sharma
Room: CB11.07.124
Phone: 9514 1835
Email: Nabin.Sharma@uts.edu.au

If you have a question about the assignment, please post it to the Canvas discussion forum for this subject so that everyone can see the response.

If serious problems are discovered, the class will be informed via an announcement on Canvas. It is your responsibility to make sure you frequently check Canvas.

Extensions and Special Consideration

Please refer to the process for special consideration.